AD Administration Points

1️⃣ Active Directory Fundamentals

1. Active Directory (AD) is a directory service used to manage users, computers, and resources.
2. AD provides centralized authentication and authorization.
3. AD stores information in a hierarchical structure.
4. AD helps administrators manage permissions and access.
5. AD uses Domain Controllers to manage authentication requests.
6. AD supports Single Sign-On (SSO).
7. AD improves security and centralized management.
8. AD stores data in a database called NTDS.dit.
9. AD uses LDAP protocol for directory access.
10. AD uses Kerberos protocol for authentication.

2️⃣ Active Directory Components

1. Domain is a logical group of users and computers.
2. Tree is a collection of one or more domains.
3. Forest is a collection of multiple domain trees.
4. Organizational Unit (OU) is used to organize objects in AD.
5. Domain Controller (DC) is a server that runs AD services.
6. Global Catalog (GC) stores searchable information about objects.
7. Schema defines the structure of AD objects.
8. Sites represent physical network locations.
9. Trusts allow access between different domains.

3️⃣ AD DS Installation and Configuration

1. AD DS role must be installed to create a Domain Controller.
2. Server must have a static IP before AD installation.
3. DCPromo promotes a server to Domain Controller (older versions).
4. New versions use Server Manager to install AD DS.
5. DNS is required for Active Directory to function properly.
6. Domain functional level determines AD features.
7. Forest functional level determines forest features.
8. SYSVOL folder stores Group Policy and scripts.
9. NTDS database stores directory information.

4️⃣ User and Group Management

1. Users are created to allow login access.
2. Groups simplify permission management.
3. Security groups are used to assign permissions.
4. Distribution groups are used for email distribution.
5. Group scope types include Domain Local, Global, and Universal.
6. Domain Local groups assign permissions within a domain.
7. Global groups contain users from the same domain.
8. Universal groups contain users from multiple domains.
9. User account properties include password policy and account lockout.
10. Disabled accounts cannot log in.

5️⃣ Group Policy

1. Group Policy is used to manage user and computer settings.
2. GPO stands for Group Policy Object.
3. GPO can be applied to Site, Domain, or OU.
4. Group Policy enforces security settings.
5. Password policy is configured through Group Policy.
6. Software can be deployed using Group Policy.
7. Scripts can be assigned using Group Policy.
8. Group Policy updates using gpupdate command.
9. Resultant Set of Policy (RSoP) shows applied policies.

6️⃣ DNS in Active Directory

1. DNS is required for AD to locate Domain Controllers.
2. AD uses SRV records in DNS.
3. Forward Lookup Zone resolves names to IP addresses.
4. Reverse Lookup Zone resolves IP to names.
5. DNS must be configured correctly for AD login.
6. AD-integrated DNS stores DNS in AD database.
7. Replication occurs automatically in AD-integrated DNS.

7️⃣ AD Authentication and Security

1. Kerberos is the default authentication protocol.
2. NTLM is used for backward compatibility.
3. Authentication verifies identity.
4. Authorization determines access level.

5. Access Control List (ACL) controls permissions.
6. Least privilege principle improves security.
7. Account lockout policy prevents brute force attacks.
8. Multi-Factor Authentication improves security.

8️⃣ Replication and Sites

1. Replication copies AD data between Domain Controllers.
2. Intrasite replication occurs within the same site.
3. Intersite replication occurs between different sites.
4. Sites reduce authentication traffic across WAN links.
5. Knowledge Consistency Checker (KCC) manages replication topology.
6. Replication ensures AD database consistency.

9️⃣ AD Maintenance and Troubleshooting

1. Backup of System State is required for AD recovery.
2. Authoritative restore restores specific AD objects.
3. Non-authoritative restore restores full AD database.
4. Event Viewer is used to troubleshoot AD issues.
5. Repadmin command checks replication health.
6. Dcdiag command checks Domain Controller health.
7. Proper DNS configuration prevents login issues.
8. Time synchronization is required for Kerberos authentication.